

Faculytics 2.0

Topic Modeling Pipeline — Full Tuning Report

Runs 001 – 013 | LaBSE + BERTopic + KeyBERTInspired

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Repo: github.com/CtrlAltElite-Devs/topic-modeling.faculytics

Executive Summary

This report documents all 13 runs of the Faculytics 2.0 multilingual topic modeling pipeline, from initial baseline through architecture pivots to production baseline selection. The pipeline discovers educator-interpretable themes from student feedback written in English, Cebuano, Tagalog, Taglish, and code-switched variants.

Final Architecture (Run 011 onwards): Multilingual preprocessing → LaBSE embeddings (Feng et al., 2022) → BERTopic with UMAP + HDBSCAN (Grootendorst, 2022; McInnes et al., 2017) → KeyBERTInspired keyword extraction.

Production Baseline (Run 013): Sentiment-gated real dataset — negative/neutral feedback always included; positive feedback included only if ≥ 10 words. Applied to UC internal evaluation portal data (34,727 raw entries → 12,297 after gating and preprocessing).

Metric Targets vs Final Results

Metric	Target	Run 011 (augmented)	Run 013 (production)	Status
Embedding Coherence	> 0.5	0.5968	0.6495	■ PASS
Topic Diversity	> 0.7	0.9474	0.9789	■ PASS
Outlier Ratio	< 20%	33.0%	52.0%	■ Data-inherent
Topic Count	10–25	19	19	■ PASS

Note: Outlier ratio is data-inherent for HDBSCAN on short multilingual text (consistent across all runs; accepted as thesis limitation).

Pipeline Architecture

1. Multilingual Preprocessing

Raw feedback undergoes a multi-stage cleaning pipeline designed for short, informal, code-switched text from Philippine students:

- Drop: Excel formula artifacts (#NAME?), entries shorter than 3 words post-cleaning, pure keyboard gibberish
- Strip: broken Unicode (■), laughter noise (hahaha/hehe/lol), repeated characters (everrr→ever), punctuation spam (???→.), URLs
- Multilingual stop words: English function words + Cebuano possessives (iyang, kanyang, among, atong) + role words (propesor, estudyante, guro, maam, sir) + Tagalog function words

No spell correction is applied — LaBSE's WordPiece tokenization provides inherent robustness to misspellings, and no Cebuano/Tagalog lexicons are available for automated correction.

2. LaBSE Embedding (Feng et al., 2022)

Language-agnostic BERT Sentence Embedding (LaBSE) encodes all feedback into a shared 768-dimensional semantic space. Trained on 109 languages including low-resource Philippine languages, LaBSE handles mid-sentence code-switching natively without requiring language detection or translation preprocessing. Embeddings are cached per dataset to avoid redundant GPU computation.

3. BERTopic Clustering (Grootendorst, 2022)

BERTopic discovers topics via two sequential steps:

- **UMAP** (McInnes et al., 2020): Reduces 768-dim LaBSE embeddings to 10 dimensions (n_neighbors=20, n_components=10, metric=cosine, random_state=42) preserving local semantic structure
- **HDBSCAN** (McInnes et al., 2017): Density-based clustering (min_cluster_size=15, min_samples=5, cluster_selection_method=eom). Documents that don't fit any cluster are assigned to noise class (-1 / outliers). nr_topics=20 triggers hierarchical merging to an educator-interpretable count.

4. KeyBERTInspired Keyword Extraction (Grootendorst, 2022 §4.2)

Replaces the default c-TF-IDF keyword extraction. KeyBERTInspired selects keywords by computing cosine similarity between candidate word embeddings and the topic centroid in LaBSE space — finding words semantically closest to the cluster's meaning rather than just the most frequent words. This is language-agnostic and eliminates the stop-word treadmill observed in Runs 001–010 (where each run revealed new frequency-dominant noise requiring manual stop-word addition).

5. Sentiment-Gated Pre-filtering (Production)

Applied before topic modeling in production (Run 013 onwards). Since Faculytics already runs sentiment analysis (Qwen 3B LoRA), the sentiment signal gates topic modeling input:

- Negative feedback → always included (primary source of actionable insights)
- Neutral feedback → always included
- Positive feedback, ≥10 words → included (substantive positive feedback)

- Positive feedback, <10 words → excluded ('awesome teacher', 'good job', '10/10')

This filter removed 35.4% of real feedback (7,086 entries) that would otherwise dominate topic clusters with generic praise noise.

Evaluation Metrics

Primary: Embedding Coherence

Average pairwise cosine similarity between LaBSE embeddings of each topic's top-10 keywords. Higher values indicate that the keywords of a topic are semantically close together in the shared embedding space — i.e., they genuinely belong to the same concept. Target: > 0.5 .

This metric was chosen over NPMI because NPMI assumes monolingual co-occurrence: semantically equivalent words in different languages (e.g., 'teacher' and 'guro') never appear in the same document, producing systematically negative NPMI scores that are a structural artifact of multilingual data, not a quality signal (Hoyle et al., 2021; Lau et al., 2014).

Secondary: Topic Diversity

Ratio of unique words across all topics' top-10 keyword lists. Low diversity indicates overlapping or redundant topics. Target: > 0.7 .

Reference Only: NPMI Coherence

Retained for completeness but not used as primary metric. Consistently 0.0 or negative across all multilingual runs — confirmed structural artifact.

Outlier Ratio

Proportion of documents assigned to HDBSCAN noise cluster (-1). Consistent at ~33% on augmented data across all runs regardless of parameter variation. Higher on real data (47–52%) due to greater linguistic diversity and shorter average entry length. Accepted as data-inherent and documented as thesis limitation.

Run-by-Run Log

Run 001 — Baseline — no preprocessing, default BERTopic params

Dataset: augmented (7,068)

Embed Coherence	Topic Diversity	Outlier Ratio	Num Topics
—	0.8013	0.4239	78

Baseline. 78 topics — far too many. No stop words. NPMI used as sole metric. c-TF-IDF keywords dominated by generic noise (maam, sir, teacher).

Run 002 — Added nr_topics=20 to force merge

Dataset: augmented (7,068)

Embed Coherence	Topic Diversity	Outlier Ratio	Num Topics
—	0.8447	0.4290	38

Added nr_topics=20 to force hierarchical merge. Down to 38 topics — improvement. Still high outlier rate. NPMI went negative — first sign of multilingual metric problem.

Run 003 — Reduced min_topic_size to 10

Dataset: augmented (7,068)

Embed Coherence	Topic Diversity	Outlier Ratio	Num Topics
—	0.8366	0.3297	82

Reduced min_topic_size to 10. Topics jumped to 82 — too granular. Confirmed nr_topics constraint is necessary.

Run 004 — Added embedding_coherence metric; UMAP/HDBSCAN tuning grid start

Dataset: augmented (7,068)

Embed Coherence	Topic Diversity	Outlier Ratio	Num Topics
0.4788	0.8474	0.3297	19

Introduced embedding_coherence as primary metric. Added UMAP/HDBSCAN grid search. First clean 19-topic result. Composite score formula established.

Run 005 — Grid: min_topic_size=20, nr_topics=15

Dataset: augmented (7,068)

Embed Coherence	Topic Diversity	Outlier Ratio	Num Topics
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0.4937	0.8071	0.3472	14
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Grid point: larger min_topic_size (20), fewer topics (15). Topic count dropped below interpretable range. Coherence slightly higher but diversity dropped.

Run 006 — Grid: min_topic_size=10, nr_topics=None (auto)

Dataset: augmented (7,068)

Embed Coherence	Topic Diversity	Outlier Ratio	Num Topics
0.4942	0.8481	0.3062	52

Grid: small clusters (min_topic_size=10), auto topic count (nr_topics=None). 52 topics — too many for educators. Composite score penalized.

Run 007 — Grid: min_topic_size=30, nr_topics=10

Dataset: augmented (7,068)

Embed Coherence	Topic Diversity	Outlier Ratio	Num Topics
0.4925	0.7667	0.3439	9

Grid: large clusters (min_topic_size=30), very few topics (10). Below minimum target of 10 interpretable topics. Coherence plateau confirmed.

Run 008 — Added role stop words; fixed embedding coherence bug (LaBSE not loaded)

Dataset: augmented (7,068)

Embed Coherence	Topic Diversity	Outlier Ratio	Num Topics
0.0000	0.8789	0.3304	19

Added role words to stop list. Fixed critical bug: LaBSE not loaded in evaluate.py → embedding_coherence was computing random vector similarities = 0.0. Bug masked all coherence improvements since Run 004.

Run 009 — Added Cebuano possessives/fillers to stop list

Dataset: augmented (7,068)

Embed Coherence	Topic Diversity	Outlier Ratio	Num Topics
0.4826	0.9053	0.3304	19

Added Cebuano possessives and fillers to stop word list. First valid coherence reading post-bug-fix: 0.4826. 11/19 topics clearly interpretable. Identified stop-word treadmill problem: c-TF-IDF is architecturally frequency-based.

Run 010 — Added English generic verbs/fillers — confirmed plateau

Dataset: augmented (7,068)

Embed Coherence	Topic Diversity	Outlier Ratio	Num Topics
0.4823	0.9053	0.3304	19

Added English generic verbs. Metrics identical to Run 009 — confirmed plateau. Stop-word expansion no longer effective. Architectural change required.

Run 011 — ARCHITECTURE PIVOT: KeyBERTInspired replaces c-TF-IDF — major quality jump

Dataset: augmented (7,068)

Embed Coherence	Topic Diversity	Outlier Ratio	Num Topics
0.5968	0.9474	0.3304	19

ARCHITECTURE PIVOT: Replaced c-TF-IDF with KeyBERTInspired representation model. Coherence: 0.4826→0.5968 (+23.7%). Diversity: 0.9053→0.9474. 18/19 topics clearly interpretable. First run to pass coherence target. AUGMENTED DATASET BASELINE LOCKED.

Run 012 — Validated Run 011 params on real UC dataset (unfiltered)

Dataset: real raw (18,559 / 34k)

Embed Coherence	Topic Diversity	Outlier Ratio	Num Topics
0.6812	0.9842	0.4776	19

Validated Run 011 architecture on real UC dataset (unfiltered, 34k→18.5k after preprocessing). Coherence and diversity higher than augmented, but outlier rate jumped to 47.8%. Topic quality degraded: Topics 0+1 near-duplicate 'good teaching' clusters, Christmas greetings topic, '10/10' rating topic, slang clusters. Root cause: real feedback is overwhelmingly short positive praise noise.

Run 013 — Sentiment-gated filter: positive <10 words excluded; production baseline

Dataset: real filtered (12,297 / 34k)

Embed Coherence	Topic Diversity	Outlier Ratio	Num Topics
0.6495	0.9789	0.5202	19

Applied sentiment-gated pre-filter: positive <10 words excluded (35.4% of real data). Filtered dataset: 12,297 docs. New topics emerged: asynchronous attendance (Topic 5), specific tardiness timing (Topic 12). Outlier rate increased to 52% (filter removed easy-to-cluster positive entries). PRODUCTION BASELINE SELECTED.

Metric Progression — All Runs

Run	Dataset	Embed Coherence	Topic Diversity	Outlier Ratio	Num Topics
001	augmented	—	0.8013	0.4239	78
002	augmented	—	0.8447	0.4290	38
003	augmented	—	0.8366	0.3297	82
004	augmented	0.4788	0.8474	0.3297	19
005	augmented	0.4937	0.8071	0.3472	14
006	augmented	0.4942	0.8481	0.3062	52
007	augmented	0.4925	0.7667	0.3439	9
008	augmented	0.0000	0.8789	0.3304	19
009	augmented	0.4826	0.9053	0.3304	19
010	augmented	0.4823	0.9053	0.3304	19
011	augmented	0.5968	0.9474	0.3304	19
012	real-raw	0.6812	0.9842	0.4776	19
013	real-filtered	0.6495	0.9789	0.5202	19

Green = augmented dataset baseline (Run 011). Blue = production baseline (Run 013).

Topic Inventory: Run 011 — Augmented Dataset Baseline (KeyBERTInspired)

#	Topic Label	Docs	Top Keywords
0	0_gigamit_ginamit_naggamit_gumagamit	142	gigamit, ginamit, naggamit, gumagamit, nag use, talakayan, use, mag aaral, uses,...
1	1_grading system_grading_inconsistent grading	78	grading system, grading, inconsistent grading, grading criteria, grade, final gr...
2	2_teaching style_paraan pagtuturo_teaching methods	46	teaching style, paraan pagtuturo, teaching methods, paagi pagtudlo, teaching app...
3	3_aaral laging_estudyanteng_mag aaral_estudyanteng nahihipan	295	aaral, laging, estudyanteng, mag aaral, estudyanteng nahihipan, laging, questio...
4	4_class schedule_unang adlaw_first day syllabus	352	class schedule, unang adlaw, first day, syllabus, first week, adlaw, first meeti...
5	5_bahin leksyon_kanamo_daling masabtan_topic	294	bahin leksyon, kanamo, daling masabtan, topic, teaching style, subject, teaching...
6	6_remind ma_email ma_announce ma_requires ma nagbigay nagpadala	136	remind ma, email ma, announce ma, ma requires, nag send, ma nagbigay, nagpadala,...
7	7_kanunay andam_kanunay adunay_laging handa halimbawa palaging	188	kanunay andam, kanunay adunay, laging handa, halimbawa palaging, palaging, kanun...
8	8_positibong feedback_walay feedback_feedback laging aming output constructive	107	positibong feedback, walay feedback, feedback laging, aming output, constructive...
9	9_walang malinaw_walang paliwanag_malinaw pamantayan	102	walang malinaw, walang paliwanag, malinaw pamantayan, without explanation, paliw...
10	10_posted portal_portal_portal ma online portal	95	posted portal, portal, portal ma, online portal, portal nag, nag post, post, sub...
11	11_ma ma_intimidating ma_inspiring ma_mastery ma approachable ma explain ma	92	ma ma, intimidating ma, inspiring ma, mastery ma, approachable ma, explain ma, m...
12	12_practical examples_world examples_real world totoong tinuod kinabuhi totoon	78	practical examples, world examples, real world, totoong, tinuod kinabuhi, totoon...
13	13_theoretical concepts_classroom practical lectures always theoretical entire	71	theoretical concepts, classroom, practical, lectures always, theoretical, entire...
14	14_subject walang subject mastery subject mastery subject magtuturo subject	59	subject walang, subject mastery, mastery subject, pagtuturo subject, subject ma,...
15	15_laging maling_laging pagkakamali_palagi laging lahat laging laging pagkiling	31	laging maling, laging pagkakamali, palagi laging, lahat laging, laging pagkiling...
16	16_lisod makigsulti_mahirap_lisod makig_lisod makigsulti mahirap lisod istorya bahin makig istorya mak	25	lisod makigsulti, mahirap, lisod makig, lisod, istorya bahin, makig istorya, mak...
17	17_office hours_hours office_available office consultation hours available	22	office hours, hours office, available office, hours, consultation hours, availab...
18	18_nagtatanggap gawa_gawa itinakdang itinatagap paraan paraan nagtatanggap nagt	17	nagtatanggap gawa, gawa itinakdang, itinatagap paraan, paraan nagtatanggap, nagt...

Topic Inventory: Run 013 — Production Baseline (Sentiment-Gated Real Data)

#	Topic Label	Docs	Top Keywords
0	0_good teaching_teaching_teach	2537	teach good teaching, teaching, teaches, teach, hes good, good, classroom, teach us, le...
1	1_thank teaching_thank god_thank	1249	good always thank teaching, thank god, thank good, always thank, hello thank, thankyou, god ...
2	2_class regularly_regularly attend	503	class regularly, regularly attend, regularly attends, attends class, class often...
3	3_learned many_learned lot_learn	3090	learned many, learned lot, learn lot, lot learned, lot learn, daghan kog, lot le...
4	4_back because_listening_back	308	back because, listening, back, listen, lessons before, sitting back, involving b...
5	5_asynchronous classes_asynchronous session	151	asynchronous classes, asynchronous session, synchronous class, asynchronous acti...
6	6_approachable attentive_instructors	144	like expert approachable attentive, instructors like, expert teach, approachable pagka, bett...
7	7_sometimes late_sometimes absent	132	late sometimes, sometimes absent, always late, late always, late sometimes, alwa...
8	8_grado gamay_grades gamay_low	122	grade low grado gamay, grades gamay, low grade, low grades, hatag grado, gamay hatag, dako...
9	9_goods raman_goods ras_goods	97	kaau goods kaau, goods ras, goods kaau, goods kaayu, goods ka, salamat goods, goods ...
10	10_ka sakin_ka akin_ambot nimo	94	amping ka sakin, ka akin, ambot nimo, amping ka, maayo kayka, buotan ka, nimo ambot, ka...
11	11_nothing comment_comments_dont	75	comment, comments dont, dont comment, ma comment, comment dont, any comm...
12	12_minutes late_30 minutes_always	53	late 30 minutes, 30 minutes, always late, 30 mins, 30mins, mins late, 20 minutes, 2...
13	13_merry christmas_happy new year	42	christmas merry christmas, happy new, new year, christmas happy, year happy, fun good, chr...
14	14_many activity_many activities_too	31	activity many activity, many activities, too activities, lot activities, activity always,...
15	15_time submit_end semester_time	24	end submitted limit, end semester, time end, submitted time, end sem, requirements, time...
16	16_approachable understandable	17	understandable approachable understandable, understandable approachable, understanding approach...
17	17_okay better_better okay_good	17	best best okay better, better okay, good best, best good, better, good, good good, wrong g...
18	18_most time_english most_speak	15	english speak time, english most, speak english, speak, speaks, most, speaking, discuss, ...

Production Baseline Recommendation

Based on 13 experimental runs, the following configuration is recommended for production deployment in the Faculytics 2.0 worker:

Component	Setting	Rationale
Embedding model	LaBSE (sentence-transformers/LaBSE)	109-language support, handles code-switching natively
Preprocessing	Multilingual stop words + noise filters	Removes Excel artifacts, role words, Cebuano/Tagalog fillers
Sentiment gate	Positive <10 words → excluded	Removes generic praise noise (35.4% of real feedback)
UMAP	n_neighbors=20, n_components=10, cosine	Preserves local semantic structure at scale
HDBSCAN	min_cluster_size=15, min_samples=5, epsilon	Stable cluster detection; eom avoids leaf-level over-splitting
Topic count	nr_topics=20	Hierarchical merge to educator-interpretable count
Keywords	KeyBERTInspired (LaBSE-backed)	Language-agnostic; immune to frequency-based stop-word treadmill
Primary metric	Embedding coherence > 0.5	Valid for multilingual data; NPMI is not
Min feedback	30 responses per faculty/semester	Below threshold: sentiment-only, no topic modeling

Known Limitations

- **Outlier rate** (~33–52%): Consistent across all parameter combinations on both datasets. Data-inherent for short multilingual text with HDBSCAN. Outlier documents still receive sentiment analysis.
- **Positive long feedback** clusters (Topics 0, 1 in Run 013): Long positive feedback still generates 'good teaching' clusters. Acceptable as dashboard insight ('majority of feedback is positive about teaching quality'). Handle at UI layer — do not surface these clusters as 'problems'.
- **Seasonal noise** (Christmas greetings, Topic 13): ~42 entries survived the word count filter by being verbose. Low document count — acceptable.
- **Feedback source**: Data is from the school's internal evaluation portal, not Moodle. Moodle provides course/student metadata; feedback data is separate.
- **No spell correction**: No Cebuano/Tagalog lexicons available. LaBSE WordPiece tokenization provides robustness. Defensible limitation for thesis.

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